FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVIII

JEE (Main)-2021

TEST DATE: 21-08-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. $v = \sqrt{gh}$

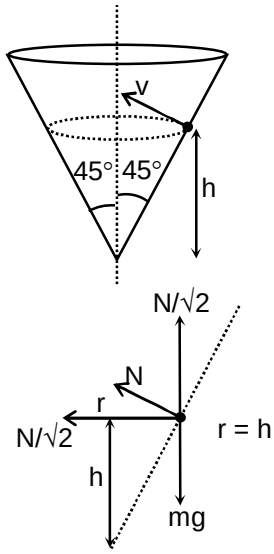
Now, when $v \to 2v = \sqrt{4gh}$, then

Towards centre $\Rightarrow \frac{N}{\sqrt{2}} = \frac{m(2v)^2}{r} = \frac{m \times 4gh}{h} = 4mg$

In vertical direction $\Rightarrow \frac{N}{\sqrt{2}} - mg = ma_r = ma_v \Rightarrow a_v = 3g$

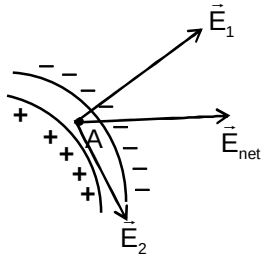
$a_H = 4g$

$a_{net} = \sqrt{a_H^2 + a_v^2} = 5g$



2. D

Sol.



3. C

Sol. Let assume that point P displaces x during the process then pulley also displaces x and the block displaces 2x.

Therefore $2x - x = \ell$ (displacement of block w.r.t. pulley)

$\Rightarrow x = \ell$

Now applying work energy theorems.

$W_{ext} + W_g = \Delta K$(of block)

$$W_{ext} - mg(2\ell) = \frac{1}{2}m\left(2\sqrt{2g\ell}\right)^2$$

$$W_{ext} = 6mg\ell$$

4. B

Sol. For diverging mirror

5. A

Sol. For Carnot's engine $\Rightarrow Q_1 = W + Q_2$

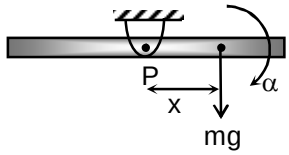
$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$$

For Carnot's refrigerator $\Rightarrow \alpha = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$

$$\frac{\eta}{\alpha} = \frac{(T_1 - T_2)^2}{T_1 T_2}$$

6. C

Sol. $\alpha = \frac{|\vec{\tau}_p|}{I} = \frac{mgx}{\frac{mL^2}{12} + mx^2}$, α is maximum at $x = \frac{L}{\sqrt{12}}$

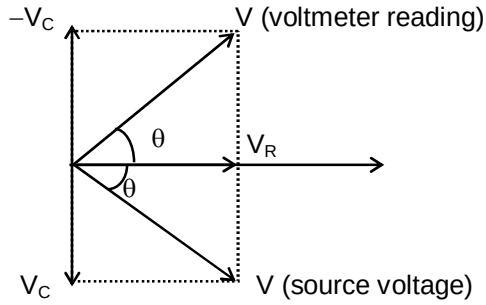


$$\alpha_{max} = \frac{\sqrt{3}g}{L}$$

7. C

Sol. Phasor diagram

$\Delta\phi = 2\theta = 2\tan^{-1}\left(\frac{3}{4}\right)$



8. A

Sol. For lens $\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{u}{v}-1=\frac{u}{f}$

$m = \frac{f}{f+u}$, $m_1 = \frac{f}{f-u_1}$ and $m_2 = \frac{f}{f-u_2}$

Two cases are possible

Case I: Both images are either real or virtual

So, $m_2 = 2m_1 \Rightarrow f + u_1 - 2u_2 = 0$ (not in option)

Case II: One image is real and other is virtual

So, $m_2 = -2m_1 \Rightarrow 3f - u_1 - 2u_2 = 0$

9. C

Sol. $\beta = \frac{\Delta I_C}{\Delta I_B} = \frac{0.98}{0.02} = 49$

$\Delta V(\text{across } R_L) = \beta \Delta I_B R_L = 49 \times 50 \times 10^{-6} \times 10^4 = 24.5 \text{ V}$

10. B

Sol. $\frac{mv^2}{r} = BeV$ and $mvr = \frac{nh}{2\pi}$

So, $r \propto \sqrt{n}$

11. A

Sol. $R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3}$

$\frac{V^2}{P_{eq}} = \frac{V^2}{P} + \frac{\frac{(2V)^2}{P} \times \frac{(3V)^2}{2P}}{\frac{(2V)^2}{P} + \frac{(3V)^2}{2P}}$

$P_{eq} = \frac{17P}{53}$

12. A

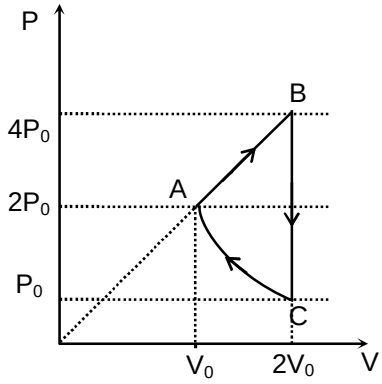
Sol. $W_{AB} = \frac{1}{2}[6P_0]V_0 = 3P_0V_0$,

$\Delta U_{AB} = nC_v\Delta T = \eta\frac{3R}{2}[T_B - T_A] = 9P_0V_0$

$Q_{AB} = 12P_0V_0$

$W_{CA} = nRT\ell n\left(\frac{V_A}{V_C}\right) = -nR\frac{P_0(2V_0)}{nR}\ell n(2) = -2P_0V_0\ell n2$

$\eta = \frac{3P_0V_0 - 2P_0V_0\ell n2}{12P_0V_0} = \frac{3 - 2\ell n2}{12}$



13. B

14. D

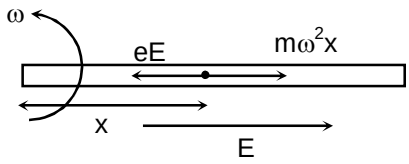
Sol. In the frame of rod, electron is in equilibrium

$m\omega^2x = eE$

$E = \frac{m\omega^2x}{e} = -\frac{dU}{dx}$

$\int dU = -\int_0^L \frac{m\omega^2x}{e}dx$

$|\Delta v| = \frac{m\omega^2L^2}{2e} = \frac{\omega^2L^2}{2\gamma}$



15. B

Sol. $Q = ms\Delta T + mg\Delta R = mS\Delta T + mgR\alpha\Delta T$

16. B

17. B

Sol. $R_1 = S\left(\frac{\ell}{100-\ell}\right), R_2 = S\left(\frac{\ell+\alpha}{100-\ell+\beta}\right)$

If balanced point is in first half (e.g. = ℓ = 40)

$R_1 = S\left(\frac{40}{60}\right), R_2 = S\left(\frac{40+1}{60+1}\right) \Rightarrow R_2 > R_1$

If balance point is in second half $\Rightarrow R_1 > R_2$

18. C

Sol. $B = \mu_0 ni = \mu_0\lambda$ (considering it as a solenoid)

19. C

Sol. $\phi = \frac{2\pi}{\lambda}(d\sin\theta)$

$\Rightarrow d = 2\lambda$

no. of maximas = 3, no. of minimas = 4

20. D

Sol. B.E. of X = 6A

B.E. of Y = 6A – 2 + 1 = 6A – 1

Because absorption of energy decreases binding energy and release of energy increases binding energy.

SECTION – B

21. 00003.00

Sol. $3m\Delta x_{cm} = m\Delta x_1 + 2m\Delta x_2$

$$3m\left(\frac{1}{2}\frac{F}{3m}t^2\right) = m(x_0 + \Delta x_2) + 2m\Delta x_2$$

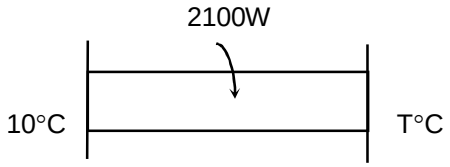
$$\Delta x_2 = \left(\frac{Ft^2}{2m} - x_0\right)\frac{1}{3} \Rightarrow \lambda = 3$$

22. 00007.00

Sol. $P = \frac{dm}{dt}S(T-10)$

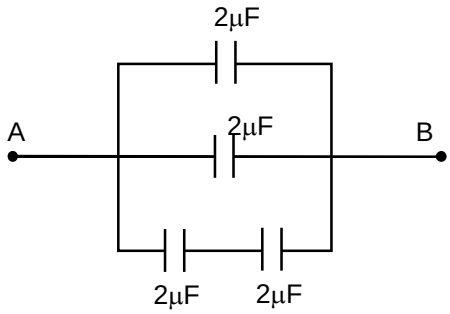
$$2100 = \frac{20}{1000}4200(T-10)\text{, } T = 35°C$$

$\Rightarrow x = 7$



23. 00004.00

Sol. $C_{eq_{AB}} = 5\mu F$



24. 00006.00

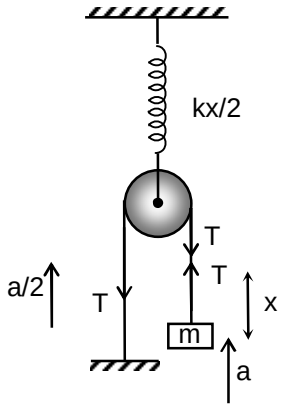
Sol. $T = ma$ ….(i)

$$\frac{kx}{2} - 2T = 2m\frac{a}{2} \quad \text{…(ii)}$$

$$\frac{kx}{2} = 3ma\text{, } \omega = \sqrt{\frac{k}{6m}}$$

$$t = \frac{T}{2} = \pi\sqrt{\frac{6m}{k}}$$

$\Rightarrow \alpha = 6$



25. 00003.00

Sol. Since the flux associated with inductors will be same just before and just after the shifting of switch. (otherwise the induced emf would be infinite).

$$i_0 L = i(L + 2L)$$

$$\frac{i_0}{i} = 3$$

26. 00005.00

Range 4.90 to 5.00

Sol. $\lambda_{cutoff} = \frac{hC}{eV_0} = \frac{1.24\times10^4}{V_0}$ Å

$$\frac{hC}{\lambda_{k\alpha}} = 13.6\left[z_{Ni}-1\right]^2\left[1-\frac{1}{4}\right]eV = 13.6\left[28-1\right]^2\frac{3}{4}eV$$

$$\lambda_{k\alpha} = \frac{1.24\times10^4}{13.6\times(27)^2\frac{3}{4}}\text{Å}$$

$\lambda_{k\alpha} - \lambda_{cutoff} = 84$ pm

$V_0 = 15$ kV

$\Rightarrow n = 5$

27. 00009.00

Sol. $H = \dfrac{k\pi R_1 R_2 \Delta T}{L}$

$$H = \frac{k\pi R(3R)\left[3T_0 - T_0\right]}{L} = \frac{k\pi(3R)\left[3R - \frac{x}{L}2R\right]\left[3T_0 - 2T_0\right]}{x}$$

28. 00008.00

Sol. $f_1 = \left(\dfrac{v - v_0}{v}\right)f,\ f_2 = \left(\dfrac{v + v_0}{v}\right)f$

Beats frequency = $\left(f_2 - f_1\right) = \dfrac{2v_0 f}{v} = 8\text{Hz}$

29. 00006.00

Sol. According to Newton's law of cooling

$$-\frac{dT}{dt} = k(T - T_S)$$

using average method

$$-\frac{(T_2 - T_1)}{t} = k\left(\frac{T_1 + T_2}{2} - T_S\right)$$

$$\frac{61-59}{4} = k\left[60-30\right],\ \ \frac{51-49}{t} = k\left[50-30\right]$$

$\Rightarrow t = 6$ minutes

30. 00000.50

Sol. $F = \eta A\dfrac{V-0}{y} + 4\eta A\dfrac{V-0}{d-y}$

$$\frac{dF}{dy} = 0$$

$$-\frac{\eta}{y^2} + \frac{4\eta}{(d-y)^2} = 0$$

$$\frac{d-y}{y} = 2$$

$$y = \frac{d}{3},\ \Rightarrow\ d_1 = \frac{d}{3},\ d_2 = \frac{2d}{3}$$

$$\frac{d_2}{d_1} = 2$$

Chemistry

PART – B

SECTION – A

31. C

Sol. PV = K

On differentiating both sides wet $PdV + VdP = 0 \Rightarrow PdV = -VdP$

$\therefore \frac{dV}{dP} = -\frac{V}{P} \Rightarrow \frac{1}{P} = \frac{-dV/dP}{V} = \beta$

Since, $\beta = \frac{1}{P}$

Inverse relation results into rectangular hyperbola. Hence, correct option is (C).

32. B

Sol. $\text{Rate} = \frac{(10-2)}{10 \text{ min}} \text{atm} = 0.8 \text{ atm m}^{-1}$

$PV = nRT \Rightarrow P = \left(\frac{n}{V}\right)RT = CRT$

$\frac{dP}{dt} = \frac{dC}{dt}(RT) \Rightarrow \frac{dC}{dt} = \frac{dP}{dt} \times \frac{1}{RT}$

$\therefore \frac{\Delta C}{\Delta t} = \frac{\Delta P}{\Delta t} \times \frac{1}{RT} = 0.8 \times \frac{1}{0.08 \times 300}$

$= 0.03 \text{ M min}^{-1}$

Hence, option (B) is correct.

33. C

Sol. $\text{Efficiency} = \frac{\Delta G^o}{\Delta H^o} \Rightarrow 0.84 = \frac{-nFE^o}{\Delta H^o}$

$0.84 == \frac{2 \times E^o \times 96500}{-285 \times 1000}$

$\therefore E^o = 1.24 \text{ V}$

34. C

Sol. Hydrolysis constant $(K_h) = \frac{1}{K_{eq}}$

$= 10^{-7}$

$\left[OH^-\right] = \sqrt{K_h C} = \sqrt{0.1 \times 10^{-7}} = 10^{-4} \text{ M}$

pOH = 4, pH = 10

Hence, (C) is the correct option.

35. C

Sol. For spontaneous process $\Delta G^o < 0$

$\Delta H - T\Delta S < 0 \Rightarrow \frac{\Delta H}{\Delta S} < T$

$\frac{-38.3 \times 1000}{-113} < T$

T > 338.93 K, i.e. 66^oC.

36. C

Sol. From Rault's law

$$P = P^{o}X_A$$

$$\therefore 0.6 = 0.8X_A, \quad X_A = \frac{3}{4}$$

$$X_B = \frac{1}{4} = 0.25$$

37. B

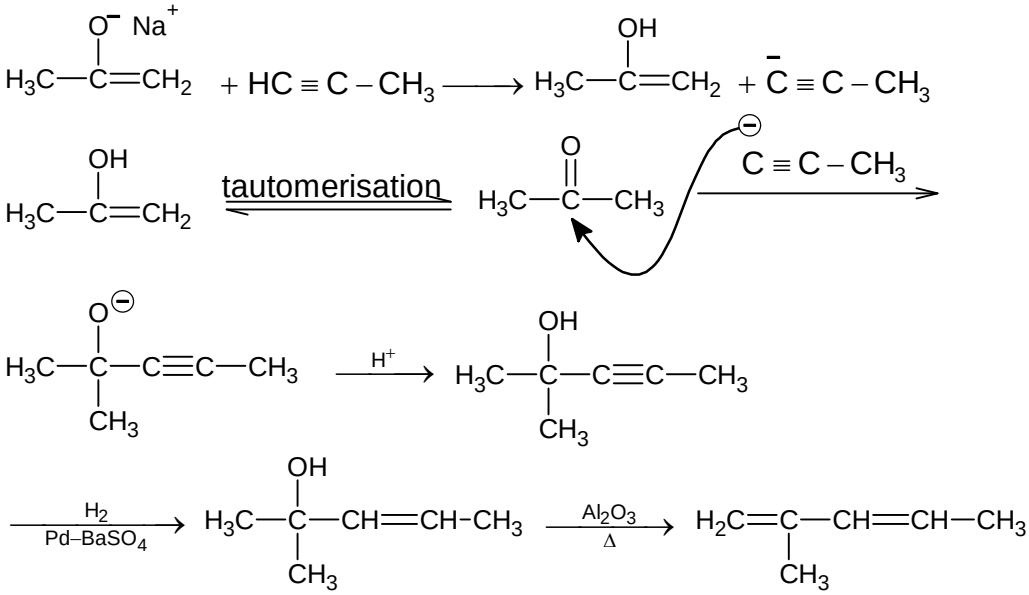
Sol. $$\lambda = \frac{h}{\sqrt{2Em}} = \frac{h}{\sqrt{2mkT}} \qquad (E = kT)$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{1200}{300}} = 2$$

$$\therefore \lambda_2 = \frac{\lambda_1}{2}$$

38. B

Sol.

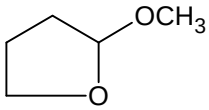


39. D

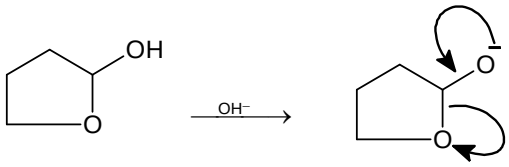
Sol. In acetylation reaction $\ddot{N}H_2$ group will be more dominating than $\ddot{O}H$ group.

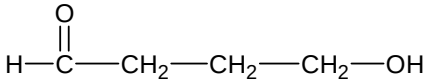
40. B

Sol.



Acetal give negative Tollen's test





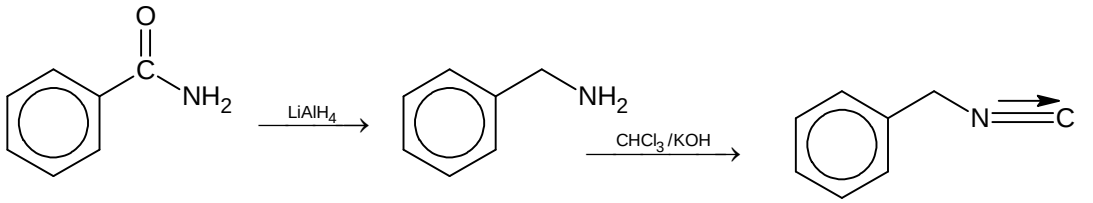
Hemiacetal give positive Tollen's test.

41. B

Sol. Ozonolysis, aldol, haloform reactions.

42. C

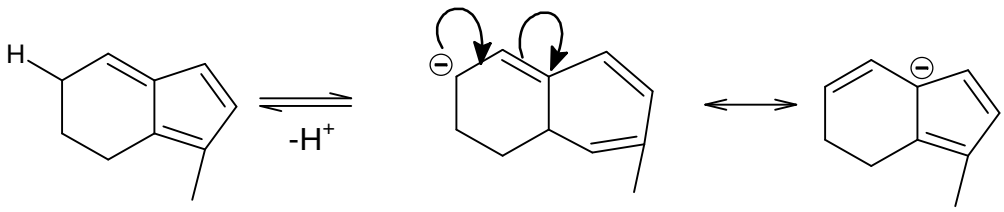
Sol.



Carbylamine (Isocyanide)

43. D

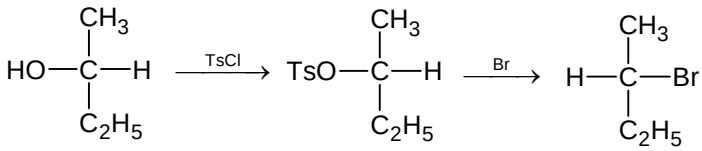
Sol.



Non-aromatic Aromatic

44. B

Sol.



45. B

Sol. $Co^{3+} = [Ar]3d^6\,4s^o\,(4 \text{ unpaired electrons})$

$Fe^{2+} = [Ar]3d^6\,4s^o\,(4 \text{ unpaired electrons})$

46. C

Sol.

Species	$XeOF_4$	XeO_2F_4	XeF_3^-	XeO_3
Lone pair(s) on central atoms	1	1	3	1

47. B

Sol. Holme's signal can be given by using $(CaC_2 + Ca_3P_2)$.

48. C

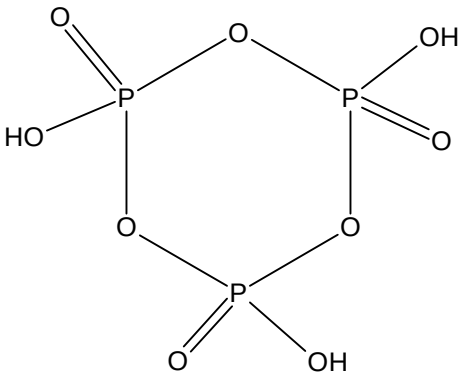
Sol. For isoelectronic species σ is same for all

$\text{Size} \propto \frac{1}{Z_{eff}}$,

Z is atomic number.

49. C

Sol.



50. A

Sol. Argentite is sulphide ore.

SECTION – B

51. 00005.99

Sol. $\Delta x = EN_B - EN_A = 2.5 - 1 = 1.5$

$\%\ \text{ionic character} = 16\Delta x + 3.5 \times (\Delta x)^2$

$\%\ \text{ionic character} = \frac{\mu_{exp}}{\mu_{th}} \times 100$

$16 \times 1.5 + 3.5 \times (1.5)^2 = \frac{\mu_{exp} \times 3.33 \times 10^{-30}}{1.6 \times 10^{-19} \times 3.92 \times 10^{-10}} \times 100$

$\mu_{exp} = 5.99$ D.

52. 00003.87

Sol. Oxidation number of central metal ion = +3

Magnetic moment $= \sqrt{3(3+2)}$

$= \sqrt{3 \times 5} = 3.87$ B.M.

53. 00001.33

Sol. $\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{C_2}{C_1}}$

$\frac{4.24}{\alpha_2} = \sqrt{\frac{0.1}{0.01}}$

$\alpha_2 = \frac{4.24}{\sqrt{10}} = 1.33$

54. 00023.20

Sol. $\kappa = C \times \frac{\ell}{a} = \frac{1}{31.6} \times 0.367$

$= 0.0116 \text{ Ohm}^{-1} \text{ cm}^{-1}$

$\Lambda_m = \kappa \times \frac{1000}{M}$

$= 0.0016 \times \frac{1000}{0.5} = 23.2 \text{ Ohm}^{-1}\text{cm}^2 \text{ mol}^{-1}$

55. 00006.25

Sol. $\lambda = \lambda_1 + \lambda_2$

$\Rightarrow \frac{0.693}{t_{1/2}} = \frac{0.693}{t_1} + \frac{0.693}{t_2} \Rightarrow \frac{1}{t_{1/2}} = \frac{t_1 + t_2}{t_1 t_2}$

$\therefore t_{1/2} = \frac{t_1 t_2}{t_1 + t_2} = \frac{4 \times 12}{4 + 12} = 3$

$N = \left(\frac{1}{2}\right)^n N_o$

$N_o = 100, \qquad n = \frac{12}{3} = 4$

$N = \left(\frac{1}{2}\right)^4 \times 100 = 6.25\%$

56. 00004.00

57. 00006.00

58. 00006.00

59. 00004.00

60. 00005.20

Sol. $H_2NCOONH_4(s) \longrightarrow CO_2(g) + 2NH_3(g)$

x moles	–	–
–	x	2x

$3x = \frac{760}{760} \times \frac{8.96}{546} \times \frac{1}{0.0821}$

$\text{Mass} = x \times M_{H_2NCOONH_4}$

$= \frac{760}{760} \times \frac{8.96}{546} \times \frac{1}{0.0821} \times \frac{1}{3} \times 78$

= 5.20 gm.

Mathematics

PART – C

SECTION – A

61. B

Sol. $\sum_{r=1}^{2n}\tan^{-1}\left(\frac{1}{2r^2}\right)=\sum_{r=1}^{2n}\left\{\tan^{-1}(2r+1)-\tan^{-1}(2r-1)\right\}=\tan^{-1}(4n+1)-\frac{\pi}{4}$

62. A

Sol. The second equation is $(x + 2)(x^2 + x + 1) = 0$
$\Rightarrow x^2 + x + 1 = 0$ and $ax^2 + bx + c = 0$ are identical
Hence, $a = b = c$

63. D

Sol. The given equation represents $y^2 = 8(x – 2)$ which is a parabola with focus (4, 0) and imaginary axis as the directrix. The two tangents from the directrix are right angle. Hence, greatest positive argument of z is $\frac{\pi}{4}$

64. C

Sol. Using ${}^nC_r=\frac{n!}{(n-r)!r!}$ we can write the given series as $\frac{30!}{10!20!}\sum_{r=0}^{10}(-1)^r\cdot 4^{10-r}\cdot\frac{10!}{r!(10-r)!}$

$\Rightarrow {}^{30}C_{10}\sum_{r=0}^{10}(-1)^r\cdot 4^{10-r}\cdot {}^{10}C_r \Rightarrow {}^{30}C_{10}(4-1)^{10}={}^{30}C_{10}\cdot 3^{10}$

65. B

Sol. $P=5^6\cdot 6^7\cdot 7^8\cdot 8^9\cdot 9^{10}.....(30)^{31}$
$=\left[(5)^6(10)^{11}(15)^{16}(20)^{21}(25)^{26}(30)^{31}\right]$ exponent of 5 in P
$= 6 + 11 + 16 + 21 + 52 + 31 = 137$

66. C

Sol. Proposition $p \to (q \vee r)$ is false. So p is true and $(q \vee r)$ is false. Therefore p is true and q and r are both false

67. A

Sol. If each item of the data is increased or decreased by some constant, the standard deviation remains unchanged

68. A

Sol. $\Delta=\begin{vmatrix}\alpha+a & b & c\\ a & \beta+b & c\\ a & b & \delta+c\end{vmatrix}=0$

Applying $R_2 \to R_2 – R_1$ and $R_3 \to R_3 – R_1$, we get $\begin{vmatrix}\alpha+a & b & c\\ -\alpha & \beta & 0\\ -\alpha & 0 & \delta\end{vmatrix}=0$

$\Rightarrow \alpha\beta c + [\beta(\alpha + a) + b\alpha]\delta = 0$ dividing by $\alpha\beta\delta$, we get $\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\delta} = -1$

69. B

Sol. $B^T = A \times (A^{-1})^T$

$B \times B^T = A^{-1}A^T \times A \times (A^{-1})^T = A^{-1} \times (A^T \times A) \times (A^{-1})^T = A^{-1}(A \times A^T) \times (A^{-1})^T$

$= (A^{-1}A) \times (A^{-1}A)^T = I \times I^T = I$

70. D

Sol. $a + b + c = 10$

Total number of non-negative solutions = ${}^{12}C_2 = 66$

Let $c = 2n \Rightarrow a + b = 10 - 2n$

$n \geq 0$, total number of non-negative solutions of above equation = 11 + 9 + 7 + 5 + 3 + 1 = 36

Probability when c is even = $\frac{36}{66} = \frac{6}{11}$; Probability when c is not even = $\frac{5}{11}$

71. D

Sol. Apply $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$ and adding, we get $2I = \int_{-1}^{1} (1+x)^{\frac{1}{2}}(1-x)^{\frac{1}{2}}\left[(1-x)+(1+x)\right]dx$

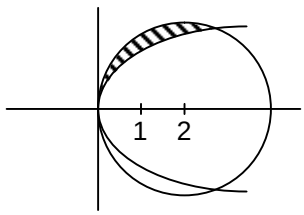
$\Rightarrow I = 2\int_0^1 \sqrt{1-x^2}dx = \frac{\pi}{2}$

72. D

Sol. $I = \frac{1}{4}\int \frac{\frac{4}{x^3} - \frac{4}{x^5}}{\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}}dx \Rightarrow I = \frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + c$

73. A

Sol. Area = $\frac{1}{4} \times \pi \times 4 - \int_0^2 \sqrt{2x}dx = \pi - \frac{8}{3}$



74. A

Sol. $(2y\,dx + x\,dy) + (xy^3\,dx + x^2y^2\,dy) = 0$ multiplying by x, we get

$(2xy\,dx + x^2\,dy) + (x^2y^3\,dx + x^3y^2\,dy) = 0$

$d(x^2y) + \frac{1}{3}d\left(x^3y^3\right) = 0 \Rightarrow x^2y + \frac{x^3y^3}{3} = c$

75. C

Sol. From LMVT, there exists atleast (n – 1) points where $f'(x) = m$

Therefore, there exist atleast (n – 2) points where $f''(x) = 0$ (using Rolle's theorem)

76. D

Sol. $\lim\limits_{n\to\infty} \cos\left(\pi\sqrt{n^2+n}\right) = \lim\limits_{n\to\infty} (-1)^n \cos\left(n\pi - \pi\sqrt{n^2+n}\right) = \lim\limits_{n\to\infty} (-1)^n \cos\left(\pi\left(n-\sqrt{n^2+n}\right)\right)$

$= \lim\limits_{n\to\infty} (-1)^n \cos\left(\dfrac{-n\pi}{n+\sqrt{n^2+n}}\right) = \lim\limits_{n\to\infty} (-1)^n \cos\left(\dfrac{n\pi}{n+\sqrt{n^2+n}}\right)$

$= \lim\limits_{n\to\infty} (-1)^n \cos\left(\dfrac{\pi}{1+\sqrt{1+\dfrac{1}{n}}}\right) = \lim\limits_{n\to\infty} (-1)^n \cos\dfrac{\pi}{2} = 0$

77. B

Sol. Let $y = mx$ be a line common to the given pair of lines, then $am^2 + 2m + 1 = 0$

and $m^2 + 2m + a = 0 \Rightarrow \dfrac{m^2}{2(1-a)} = \dfrac{m}{a^2-1} = \dfrac{1}{2(1-a)} \Rightarrow m^2 = 1$ and $m = -\dfrac{(a+1)}{2}$

$\Rightarrow (a+1)^2 = 4 \Rightarrow a = 1$ or -3

But for $a = 1$, the two pairs have both the lines common

So, $a = -3$ and the slope m of the line common to both the pairs is 1

Now, $x^2 + 2xy + ay^2 = x^2 + 2xy - 3y^2 = (x - y)(x + 3y)$

and $ax^2 + 2xy + y^2 = -3x^2 + 2xy + y^2 = -(x - y)(3x + y)$

So, the equation of required lines is $(x + 3y)(3x + y) = 0 \Rightarrow 3x^2 + 10xy + 3y^2 = 0$

78. C

Sol. Normal at $P(x_1, y_1) = \left(at_1^2, 2at_1\right)$ meets the parabola at $R(at^2, 2at)$

Similarly, normal at $Q(x_2, y_2) \equiv \left(at_2^2, 2at_2\right)$ meets the parabola at R $(at^2, 2at)$

$\Rightarrow t = -t_1 - \dfrac{2}{t_1} = -t_2 - \dfrac{2}{t_1} \Rightarrow t_1t_2 = 2$ given $x_1 + x_2 = 4$

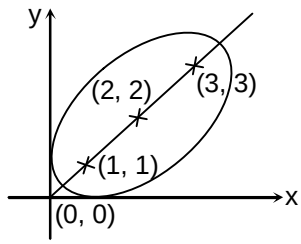
$t_1^2 + t_2^2 = 4 \Rightarrow (t_1 + t_2)^2 = 8 \Rightarrow |t_1 + t_2| = 2\sqrt{2} \Rightarrow |y_1 + y_2| = 4\sqrt{2}$

79. D

Sol. Since, x-axis, and y-axis are perpendicular tangents to the ellipse (0, 0) lies on the director circle and mid-point of foci (2, 2) is centre of circle.

Hence, radius = $2\sqrt{2}$

$\Rightarrow$ Area = 8π units



80. D

Sol. For $xy = c^2$; $x\dfrac{dy}{dx} + y = 0$ replacing $\dfrac{dy}{dx}$ by $-\dfrac{dx}{dy}$, we have $-x\dfrac{dx}{dy} + y = 0$

$\Rightarrow y\,dy - x\,dx = 0$ integrating, we get $x^2 - y^2 = k$ which represents family of rectangular hyperbola

SECTION – B

81. 00001.00

Sol. Since both the circles passes through (0, 0) the equation of tangent at (0, 0) on both the circle will be same tangent at (0, 0) to circle (1)

$(3 + \sin\beta)x + (2\cos\alpha)y = 0$

tangent at (0, 0) to circle (2)

$(2\cos\alpha)x + 2cy = 0$

$\Rightarrow$ both the equations are identical hence, $\frac{3+\sin\beta}{2\cos\alpha} = \frac{2\cos\alpha}{2c}$

$\Rightarrow c = \frac{2\cos^2\alpha}{3+\sin\beta}$ for maximum value of c, $\alpha = 0$, and $\sin\beta = -1$

$\Rightarrow$ c (max) = 1

82. 00000.50

Sol. $f(\theta) = \frac{(\cos\theta-\sin\theta)^2+\cos^2\theta-\sin^2\theta}{2(\cos\theta-\sin\theta)(\cos\theta+\sin\theta)} = \frac{1}{1+\tan\theta}$

Now, $f(11°)\cdot f(34°) = \frac{1}{1+\tan 11°}\times\frac{1}{\tan(45°-11°)}$

$\Rightarrow \frac{1}{1+\tan 11°}\times\frac{1+\tan 11°}{2} = \frac{1}{2} = 0.5$

83. 00007.00

Sol. Equation reduces to $(\cos x + \cos 4x) + (\cos 2x + \cos 3x) = 0$

$\Rightarrow 2\cos\frac{5x}{2}\left[2\cos x\cos\frac{x}{2}\right] = 0 \Rightarrow \cos\frac{5x}{2} = 0$, or $\cos x = 0$ or $\cos\frac{x}{2} = 0$

$\Rightarrow x = \frac{(2n+1)\pi}{5}$ or $x = \frac{(2n+1)\pi}{2}$ or $x = (2n+1)\pi$, $n \in z$

$\Rightarrow x = \left\{\frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}, \frac{\pi}{2}, \frac{3\pi}{2}\right\}$. Hence, number of solution 7

84. 00047.00

Sol. $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{4} = \tan^{-1}\frac{7}{11}$

$\tan^{-1}\frac{7}{11} + \tan^{-1}\left(\frac{1}{5}\right) = \tan^{-1}\frac{23}{24}$

$\tan^{-1}\frac{1}{n} = \tan^{-1}1 - \tan^{-1}\frac{23}{24} = \tan^{-1}\left(\frac{1}{47}\right) \Rightarrow n = 47$

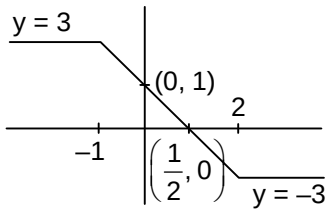
85. 00005.00

Sol. $f(x) = |x-2| - |x+1|$

$$f(x) = \begin{cases} 3 & ; \ \forall\ x < -1 \\ 1-2x & ; \ \forall\ -1 \le x \le 2 \\ -3 & ; \ \forall\ x > 2 \end{cases}$$

$\Rightarrow k \in (-3, 3)$

Hence, integral values = 5



86. 00256.00

Sol. Let the plane meets axis at A(a, 0, 0), B(0, b, 0) and C(0, 0, c)

$\Rightarrow$ equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

It's distance from origin = $\dfrac{1}{\sqrt{\dfrac{1}{a^2}+\dfrac{1}{b^2}+\dfrac{1}{c^2}}} = p$ (given)

$\Rightarrow \dfrac{1}{a^2}+\dfrac{1}{b^2}+\dfrac{1}{c^2}=\dfrac{1}{p^2}$ ..... (1)

Also, centroid $\equiv (x, y, z) \equiv \left(\dfrac{a}{4}, \dfrac{b}{4}, \dfrac{c}{4}\right)$

$\Rightarrow a = 4x, b = 4y, c = 4z$ ..... (2)

$\Rightarrow$ From (1) and (2), we get $\dfrac{1}{x^2}+\dfrac{1}{y^2}+\dfrac{1}{z^2}=\dfrac{16}{p^2}=\dfrac{\lambda}{p^2}$

$\Rightarrow \lambda^2 = 256$

87. 00000.71

Sol. $(\text{Volume})^2 = \left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{b}\cdot\vec{a} & \vec{c}\cdot\vec{a} \\ \vec{a}\cdot\vec{b} & \vec{b}\cdot\vec{b} & \vec{c}\cdot\vec{b} \\ \vec{a}\cdot\vec{c} & \vec{b}\cdot\vec{c} & \vec{c}\cdot\vec{c} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix} = \frac{1}{2}$

$\Rightarrow$ Volume $= \dfrac{1}{\sqrt{2}} = 0.71$ units

88. 00000.50

Sol. Let $L = \lim\limits_{n\to\infty}\left(\prod\limits_{r=1}^{n}\left(1+\dfrac{n}{r}\right)^{\frac{r}{n}}\right)^{\frac{1}{n}}$, then $\log_e L = \lim\limits_{n\to\infty}\dfrac{1}{n}\sum\limits_{r=1}^{n}\dfrac{r}{n}\ln\left(1+\dfrac{n}{r}\right) = \int\limits_0^1 x\ln\left(1+\dfrac{1}{x}\right)dx = \dfrac{1}{2}$

89. 00520.00

Sol. Here, f(1) = 3, f(2) = 7, f(3) = 12 and so on

$S_{12} = 3 + 7 + 12 + ... + t_n$

$S_{12} = 3 + 7 + ... + t_{n-1} + t_n$

On subtracting $t_n = 3 + 4 + 5 + ...$ upto n terms $t_n = \dfrac{n^2+5n}{2}$

$S_n = \sum t_n = \dfrac{n(n+1)(n+8)}{6} \Rightarrow S_{12} = 520$

90. 00008.00

Sol. $t_2 = -t_1 - \dfrac{2}{t_1} \Rightarrow t_1^2 + t_1t_2 + 2 = 0 \Rightarrow D \geq 0 \Rightarrow t_2^2 \geq 8$. Hence, least integral value of t_2^2 is 8